

Chapter 2

BJT Basics II

Saturation Mode, PNP Transistors & Biasing Fundamentals

*Based on Razavi, Fundamentals of Microelectronics, 2nd Ed.
Chapters 4.5–4.6 (Saturation & PNP) and Chapter 5.1–5.2 (Biasing)*

With insights from Razavi's BJT YouTube Lectures (15–20)

Topics Covered:

- Review of Chapter 1 concepts (summary diagram)
- Operation in Saturation Mode (soft vs. deep)
- The PNP Transistor (structure, operation, equations)
- PNP Small-Signal Model
- DC Operating Point Analysis
- Biasing Techniques: R_B bias, voltage divider, emitter degeneration
- Separating DC Analysis from Small-Signal Analysis

Chapter 2: Quick Reference Formula Sheet

Saturation Mode

Name	Formula / Condition	Notes
Active mode	$V_{BE} > 0, V_{BC} < 0$	Normal amplification
Edge of saturation	$V_{CE} = V_{BE} \ (V_{BC} = 0)$	Boundary condition
Soft saturation	$V_{BC} < 400 \text{ mV}$	I_C still $\approx I_S e^{V_{BE}/V_T}$
Deep saturation	$V_{CE,sat} \approx 200 \text{ mV}$	$\beta_{\text{forced}} \ll \beta$

PNP Transistor (Active Mode)

Name	Formula	Notes
I_C (PNP)	$I_C = I_S \exp(V_{EB}/V_T)$	Note: V_{EB} , not V_{BE}
I_B (PNP)	$I_B = \frac{I_S}{\beta} \exp(V_{EB}/V_T)$	Same β relation
I_E (PNP)	$I_E = \frac{\beta+1}{\beta} I_S \exp(V_{EB}/V_T)$	KCL still holds
Active condition	$V_{EB} > 0, V_{CB} > 0$	$V_C < V_B$ for PNP
Early effect	$I_C = I_S e^{V_{EB}/V_T} (1 + V_{EC}/V_A)$	Same form

Biasing Equations

Topology	Key Equation	Notes
R_B bias	$I_C = \beta \frac{V_{CC} - V_{BE}}{R_B}$	Sensitive to β !
Voltage divider	$V_X = \frac{R_2}{R_1 + R_2} V_{CC}$	β -independent if $I_B \ll I_{R2}$
With R_E	$I_E \approx \frac{V_X - V_{BE}}{R_E}$	Very robust
Thevenin	$V_{Thev} = \frac{R_2}{R_1 + R_2} V_{CC}, \ R_{Thev} = R_1 \parallel R_2$	Exact analysis

1 Review: Where We Left Off

In Chapter 1, we established the BJT as a **voltage-controlled current source** operating in the forward active mode. We derived:

- $I_C = I_S \exp(V_{BE}/V_T)$ — collector current depends exponentially on V_{BE}
- $I_B = I_C/\beta$, $I_E = I_C + I_B$ — terminal current relationships
- $g_m = I_C/V_T$ — transconductance (how “strong” the BJT amplifies)
- $r_\pi = \beta/g_m$ — small-signal input resistance
- $r_O = V_A/I_C$ — output resistance from the Early effect

Figure 4.32 from Razavi provides a masterful one-page summary of everything from Chapter 1:

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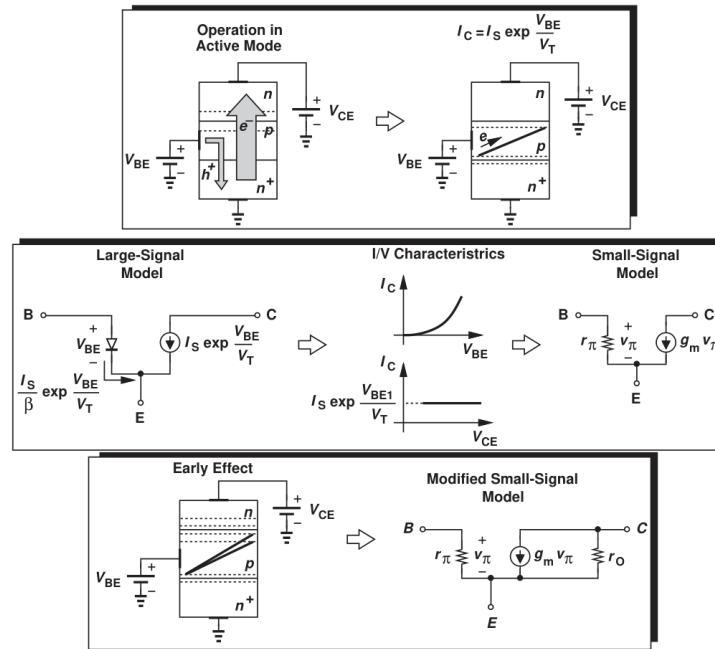


Figure 4.32 Summary of concepts studied thus far.

4.5 OPERATION OF BIPOLAR TRANSISTOR

Figure 1: Figure 4.32 from Razavi — Complete summary: physical operation, large-signal model, I/V characteristics, small-signal model, Early effect, and modified model with r_O .

EXAM TIP

Before every exam, redraw Figure 4.32 from memory. If you can reproduce all six panels (physical cross-section, large-signal model, I/V curves, basic small-signal model, Early effect mechanism, and modified small-signal model with r_O), you understand the BJT fundamentals.

This chapter answers three critical questions that Chapter 1 left open:

1. What happens when V_{CE} drops below V_{BE} ? (Saturation mode)
2. Can we flip all the polarities and make a “PNP” transistor? (Yes, and it is essential)
3. How do we actually *set* the DC operating point in a real circuit? (Biasing)

2 Operation in the Saturation Region

2.1 Physical Intuition: What Goes Wrong?

Recall from Chapter 1 that active mode requires the base-collector junction to be **reverse-biased** ($V_{BC} < 0$). This ensures that the collector “collects” electrons without injecting any carriers back into the base.

Now **imagine we lower V_{CE}** (e.g., by increasing R_C so the voltage drop $I_C R_C$ consumes most of V_{CC}). Eventually V_C drops to V_B , making $V_{BC} = 0$. If V_C drops *below* V_B , the base-collector junction becomes **forward-biased**. Now *both* junctions are forward-biased simultaneously.

CONCEPT: Why Saturation Kills Amplification

When the BC junction is forward-biased, it injects holes from the base into the collector. These holes flow in the *opposite* direction to the electron current we want. The net collector current **decreases**, and the transistor loses its behavior as a controlled current source. It becomes more like a short circuit ($V_{CE} \approx 0.2$ V) than an amplifier.

4.5 Operation of Bipolar Transistor in Saturation Mode 151

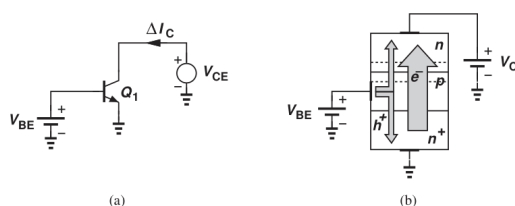


Figure 4.33 (a) Bipolar transistor with forward-biased base-collector junction, (b) flow of holes to collector.

Figure 2: Figure 4.33 from Razavi — (a) Circuit entering saturation as V_{CE} drops, (b) physical picture: holes now flow from base into collector through the forward-biased BC junction.

LECTURE INSIGHT (Lecture 17)

Razavi explains in Lecture 17: “When both junctions are forward-biased, you essentially have two diodes fighting each other. The base-emitter diode tries to push electrons from E to C, while the base-collector diode tries to push electrons from C to B. The net current drops, and the transistor is no longer a good current source. This is saturation.”

2.2 Soft Saturation vs. Deep Saturation

There is a **spectrum** of saturation severity:

Soft vs. Deep Saturation

Property	Soft Saturation	Deep Saturation
V_{BC}	$0 < V_{BC} < 400 \text{ mV}$	$V_{BC} > 400 \text{ mV}$
V_{CE}	$V_{BE} - 400\text{mV} < V_{CE} < V_{BE}$	$V_{CE} \approx V_{CE,sat} \approx 200 \text{ mV}$
I_C	Still $\approx I_S e^{V_{BE}/V_T}$	Much less than $I_S e^{V_{BE}/V_T}$
β_{forced}	Close to nominal β	$\ll \beta$ (can be 5–20)
Behavior	Marginally acceptable	Transistor $\approx$ ON switch
Speed	Somewhat degraded	Severely degraded

The **400 mV rule** is a practical boundary: a forward-biased diode carrying negligible current has $V_{BC} < 400 \text{ mV}$, so the BC junction's effect is minimal. As a design rule:

■ **KEY RESULT:**

Active-mode guarantee: $V_{CE} > V_{BE} - 400 \text{ mV}$ (soft saturation acceptable).
Strict active mode: $V_{CE} > V_{BE}$ (i.e., $V_{BC} < 0$).

2.3 The Saturation Model

In saturation, we model the transistor with **two forward-biased diodes**:

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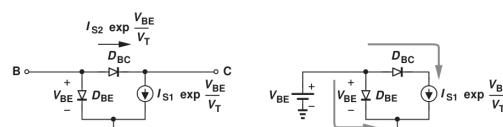


Figure 3: Figure 4.34 from Razavi — (a) Large-signal model with both diodes forward-biased, showing the BC diode D_{BC} fighting the controlled source. (b) Open-collector case.

The I/V characteristics clearly show the three regions:

is used because increasing the base current in this region of operation leads to a drop in the collector current.

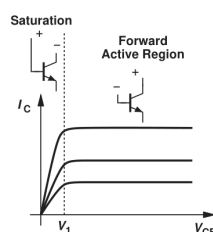


Figure 4.35 Transistor I/V characteristics in different regions of operation.

In addition to a drop in β , the speed of bipolar transistors also degrades

Figure 4: Figure 4.35 from Razavi — I_C vs. V_{CE} showing saturation (left of V_1) and forward active (right of V_1) regions. In saturation, increasing I_B barely changes I_C .

In deep saturation, the transistor is modeled simply as two batteries:



Figure 5: Figure 4.37 from Razavi — Deep saturation model: $V_{BE} \approx 800$ mV, $V_{CE} \approx 200$ mV. The transistor acts as an ON switch.

EXAM TIP

On exams, if a problem says “the transistor is in deep saturation,” immediately write $V_{CE,sat} = 0.2$ V and $V_{BE} = 0.8$ V. You do NOT use $I_C = I_S e^{V_{BE}/V_T}$ — instead, I_C is determined by the external circuit: $I_C = (V_{CC} - V_{CE,sat})/R_C$. The base current is then $I_B = I_C/\beta_{\text{forced}}$ where $\beta_{\text{forced}} < \beta$.

Example 4.16: Saturation Boundary (Razavi)

Problem: For the circuit of Fig. 4.36, determine the relationship between R_C and V_{CC} that guarantees operation in soft saturation or active region.

4.6 The PNP Transistor 153

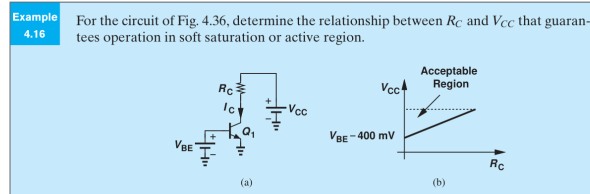


Figure 4.36 from Razavi

Solution:

In soft saturation, I_C is still $\approx I_S \exp(V_{BE}/V_T)$. The collector voltage must not fall below the base voltage by more than 400 mV:

$$V_{CC} - R_C I_C \geq V_{BE} - 400 \text{ mV}$$

Thus:

$$V_{CC} \geq I_C R_C + (V_{BE} - 400 \text{ mV})$$

For a given R_C , V_{CC} must be large enough. Alternatively, for a given V_{CC} , R_C cannot be too large.

3 The PNP Transistor

3.1 Why Do We Need PNP?

The *nnp* transistor can only “pull down” — its collector current flows downward (into the collector). Many circuit topologies require a complementary device that “pulls up.” The **pn**p transistor fills this role. Together, npn and pnp form **complementary pairs** that are essential for push-pull output stages, current mirrors, and many amplifier topologies.

3.2 Structure and Operation

The *pnp* transistor is the “mirror image” of the *nnp*: the emitter and collector are *p*-type, and the base is *n*-type. The majority carriers are now **holes** instead of electrons.

Active mode conditions for PNP:

- Base-emitter junction **forward-biased**: $V_{EB} > 0$ (i.e., $V_{BE} < 0$, emitter is more positive than base)
- Base-collector junction **reverse-biased**: $V_{CB} > 0$ (i.e., collector is more *negative* than base)

Figure 4.38(a) shows the structure of a *pnp* transistor, emphasizing that the emitter is heavily doped. As with the *nnp* counterpart, operation in the active region requires forward-biasing the base-emitter junction and reverse-biasing the collector junction. Thus, $V_{BE} < 0$ and $V_{BC} > 0$. Under this condition, majority carriers in the emitter (holes) are injected into the base and swept away into the collector. Also, a linear profile of holes is formed in the base region to allow diffusion. A small number of base majority carriers (electrons) are injected into the emitter or recombined with the holes in the base region, thus creating the base current. Figure 4.38(b) illustrates the flow of the carriers. All of the operation principles and equations described for *nnp* transistors apply to *pnp* devices as well.

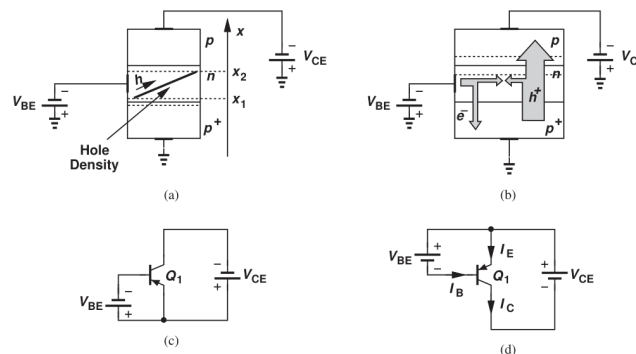


Figure 4.38 (a) Structure of *pnp* transistor, (b) current flow in *pnp* transistor, (c) proper biasing, (d) more intuitive view of (c).

Figure 4.38(c) depicts the symbol of the *pnp* transistor along with constant voltage sources that bias the device in the active region. In contrast to the biasing of the *nnp*

Figure 6: Figure 4.38 from Razavi — (a) PNP physical structure with hole density profile, (b) electron current flow for comparison, (c) circuit symbol with proper biasing, (d) redrawn with conventional “positive supply on top” orientation.

CONCEPT: The PNP “Flip Everything” Rule

To convert any *nnp* equation or condition to *pnp*:

1. Replace V_{BE} with V_{EB} (flip the subscripts)

2. Replace V_{CE} with V_{EC}
3. Reverse all current arrow directions (current enters through emitter)
4. The active mode condition becomes: $V_{EB} > 0$ and $V_C < V_B$ (collector lower than base)

Everything else — β , g_m , r_π , r_O — has the same definition and same formula.

3.3 PNP Current Equations

DERIVATION: PNP Terminal Currents

The derivation follows identically to the npn case but with holes as the majority carriers:

$$I_C = I_S \exp\left(\frac{V_{EB}}{V_T}\right) \quad (1)$$

$$I_B = \frac{I_S}{\beta} \exp\left(\frac{V_{EB}}{V_T}\right) = \frac{I_C}{\beta} \quad (2)$$

$$I_E = \frac{\beta + 1}{\beta} I_S \exp\left(\frac{V_{EB}}{V_T}\right) = I_C + I_B \quad (3)$$

Including the Early effect:

$$I_C = I_S \exp\left(\frac{V_{EB}}{V_T}\right) \left(1 + \frac{V_{EC}}{V_A}\right)$$

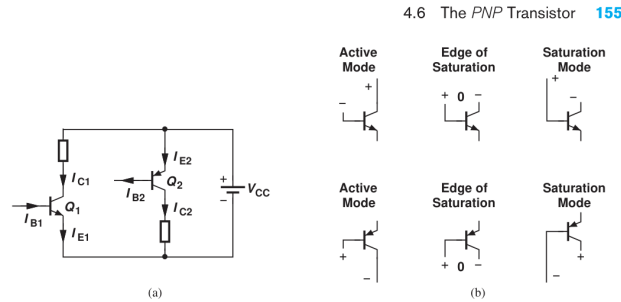


Figure 4.39 (a) Voltage and current polarities in *npn* and *pn*p transistors, (b) illustration of active and saturation regions.

Figure 7: Figure 4.39 from Razavi — (a) NPN and PNP branches with current directions, (b) active mode and saturation mode conditions for both transistor types. Note the reversed polarity convention.

3.4 PNP Large-Signal and Small-Signal Models

The large-signal model of the PNP is identical in topology to the NPN, but with reversed polarity conventions:

The small-signal model of the PNP is **identical** to the NPN model:

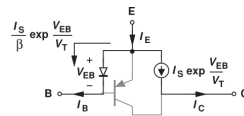
Figure 4.40 Large-signal model of *pn*p transistor.

Figure 8: Figure 4.40 from Razavi — Large-signal model of PNP transistor.

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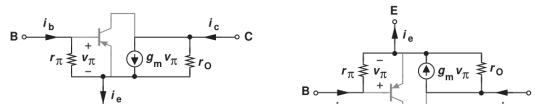


Figure 9: Figure 4.43 from Razavi — (a) PNP small-signal model, (b) redrawn in more intuitive orientation. Note: the small-signal model is *exactly* the same as NPN.

KEY RESULT:

The PNP and NPN small-signal models are identical.

$$g_m = I_C/V_T, \quad r_\pi = \beta/g_m, \quad r_o = V_A/I_C$$

The only difference is in the DC analysis (voltage/current polarities).

LECTURE INSIGHT (Lecture 18)

Razavi emphasizes in Lecture 18: “Students often panic when they see a PNP transistor, but the small-signal model is *exactly the same*. The PNP only adds complexity in the DC bias analysis, where you must be careful about voltage polarities. Once you have I_C , computing g_m and r_π is identical to the NPN case.”

Example 4.17: PNP Simple Stage (Razavi)

Problem: Determine the terminal currents of Q_1 in Fig. 4.41 and verify active mode. $I_S = 2 \times 10^{-16}$ A, $\beta = 50$, $V_A = \infty$.

In the circuit shown in Fig. 4.41, determine the terminal currents of Q_1 in the forward active region. Assume $I_S = 2 \times 10^{-16}$ A and $\beta =$

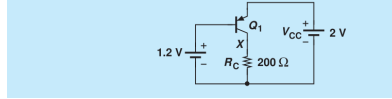


Figure 4.41 from Razavi — PNP stage with $V_{CC} = 2$ V, supply at base = 1.2 V, $R_C = 200 \Omega$

Solution:

1. Compute V_{EB} : $V_{EB} = V_{CC} - V_B = 2 - 1.2 = \mathbf{0.8}$ V
2. Collector current: $I_C = I_S \exp(V_{EB}/V_T) = (2 \times 10^{-16}) \exp(800/26) = \mathbf{4.61}$ mA
3. Base current: $I_B = I_C/\beta = 4.61 \text{ mA}/50 = \mathbf{92.2}$ μ A
4. Emitter current: $I_E = I_C + I_B = \mathbf{4.70}$ mA
5. Collector voltage: $V_X = R_C I_C = 200 \times 4.61 \text{ m} = \mathbf{0.922}$ V
6. Active mode check: $V_C = 0.922$ V, $V_B = 1.2$ V. Since $V_C < V_B$ for PNP, **active mode is confirmed.**

Example 4.18: PNP with Small-Signal Input (Razavi)

Problem: In Fig. 4.42, V_{in} is a microphone signal. Find V_{out} for $V_{in} = 0$ and $V_{in} = +5$ mV. $I_S = 1.5 \times 10^{-16}$ A, $R_C = 300 \Omega$, $V_{CC} = 2.5$ V.

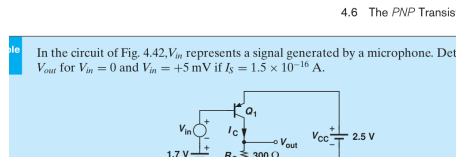


Figure 4.42 from Razavi

Solution:

For $V_{in} = 0$: $V_{EB} = 2.5 - 1.7 = 0.8$ V

$I_C = I_S \exp(V_{EB}/V_T) = \mathbf{3.46}$ mA

$V_{out} = R_C I_C = 300 \times 3.46 \text{ m} = \mathbf{1.038}$ V

For $V_{in} = +5$ mV: $V_{EB} = 800 - 5 = 795$ mV (base voltage rises $\rightarrow V_{EB}$ decreases)

$I_C = 1.5 \times 10^{-16} \exp(795/26) = \mathbf{2.85}$ mA

$V_{out} = 300 \times 2.85 \text{ m} = \mathbf{0.856}$ V

Key insight: A 5 mV change in V_{in} causes a $1.038 - 0.856 = 182$ mV change in V_{out} . The voltage gain is ≈ 36.4 .

4 DC Operating Point Analysis and Design

4.1 The Fundamental Bias Problem

Every BJT amplifier requires a **DC operating point** (also called the “Q-point” or “bias point”). This is the set of DC voltages and currents (I_C , V_{CE} , I_B) that position the transistor in the active region *before* any signal is applied.

CONCEPT: Why Biasing Matters

Without proper biasing:

- The transistor may be in cutoff (no amplification)
- The transistor may be in saturation (no amplification)
- The amplifier gain may be unpredictable (depends on unreliable β)
- Signal clipping occurs if the Q-point allows insufficient “swing room”

A well-designed bias network must:

1. Set I_C to the desired value
2. Ensure active mode ($V_{CE} > V_{BE}$ for NPN)
3. Be **insensitive to β variations** (the most important criterion!)

LECTURE INSIGHT (Lecture 19)

Razavi opens Lecture 19 with: “The whole point of biasing is to establish a *known, stable* collector current. Everything in the small-signal world — g_m , r_π , r_O , gain — depends on I_C . If I_C is wrong or unpredictable, the amplifier is useless. So biasing is not optional — it is the foundation of every amplifier.”

4.2 DC vs. Small-Signal Analysis: The Separation Principle

DC Analysis vs. Small-Signal Analysis — THE SEPARATION PRINCIPLE

	DC (Bias) Analysis	Small-Signal (AC) Analysis
Purpose	Find I_C , V_{CE} , I_B	Find gain, impedances
Variables	Uppercase: I_C , V_{BE}	Lowercase: i_c , v_{be} (v_π)
Transistor model	Exponential: $I_C = I_S e^{V_{BE}/V_T}$	Linear: $i_c = g_m v_\pi$
Signal sources	Set to zero (shorted/opened)	Active
Capacitors	Open circuits	Short circuits (at signal freq.)
V_{CC}	Active (provides current)	AC ground (shorted to GND)

Procedure:

1. **Step 1 (DC):** Kill all AC sources. Solve for I_C , V_{CE} , I_B . Verify active mode.
2. **Step 2 (Small-signal):** Compute $g_m = I_C/V_T$, $r_\pi = \beta/g_m$, $r_O = V_A/I_C$.

3. **Step 3 (Small-signal):** Replace the transistor with its small-signal model. Short V_{CC} to ground. Solve for gain.

EXAM TIP

NEVER mix DC and AC quantities in the same equation. If you find yourself writing “ $I_C + g_m v_\pi$ ” in a single equation, you have made a fundamental error. DC analysis gives you the bias point; small-signal analysis gives you gain and impedances around that point. They are solved sequentially, never simultaneously.

5 Biasing Techniques

5.1 Technique 1: Simple R_B Bias (The Bad Approach)

The simplest bias network uses a single resistor R_B from V_{CC} to the base:

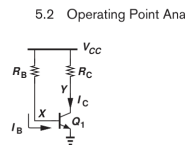


Figure 5.13 Use of base resistance for base current path.

Figure 10: Figure 5.13 from Razavi — Simple R_B biasing. The base current flows through R_B .

DERIVATION: R_B Bias Analysis

Apply KVL around the base-emitter loop:

$$R_B I_B + V_{BE} = V_{CC}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Since $I_C = \beta I_B$:

$$I_C = \beta \cdot \frac{V_{CC} - V_{BE}}{R_B}$$

The collector-emitter voltage:

$$V_{CE} = V_{CC} - R_C I_C = V_{CC} - \beta \frac{V_{CC} - V_{BE}}{R_B} R_C$$

COMMON MISTAKES

- **FATAL FLAW:** $I_C = \beta(V_{CC} - V_{BE})/R_B$. Since β varies from 50 to 200 across devices, I_C can vary by a factor of **4x**! This is **completely unacceptable** for amplifier design.
- **Temperature sensitivity:** V_{BE} decreases by ~ 2 mV/ $^{\circ}$ C. For large R_B , this has less effect, but the β sensitivity alone disqualifies this topology.
- **Bottom line:** Simple R_B bias is **never used in practice** for analog circuits. It is only useful for digital switching applications where the transistor is deliberately driven into saturation.

5.2 Technique 2: Resistive Voltage Divider Bias

To eliminate β dependence, we **fix V_{BE} directly** using a voltage divider:

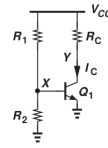
Figure 5.15 Use of resistive divider to define V_{BE} .

Figure 11: Figure 5.15 from Razavi — Resistive divider sets V_{BE} directly. The base voltage $V_X = \frac{R_2}{R_1 + R_2} V_{CC}$, which is independent of β .

DERIVATION: Voltage Divider Bias

If the base current is negligible compared to the divider current ($I_B \ll V_{CC}/(R_1 + R_2)$), then:

$$V_X = V_{BE} = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$I_C = I_S \exp\left(\frac{V_X}{V_T}\right)$$

This result is **independent of β** ! The collector current depends only on V_{CC} , R_1 , R_2 , I_S , and V_T .

Example 5.8: Voltage Divider Bias (Razavi)

Problem: Determine I_C in Fig. 5.16 if $I_S = 10^{-17}$ A, $\beta = 100$. Verify active mode and that I_B is negligible.

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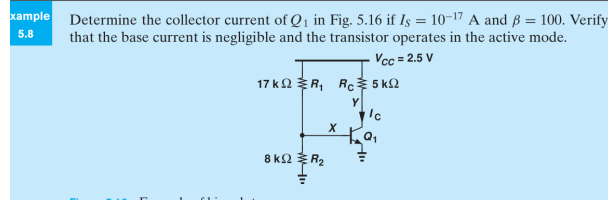


Figure 5.16 Example of biased stage.

Figure 5.16 from Razavi — $R_1 = 17 \text{ k}\Omega$, $R_2 = 8 \text{ k}\Omega$, $R_C = 5 \text{ k}\Omega$, $V_{CC} = 2.5 \text{ V}$

Solution:

1. Divider voltage: $V_X = \frac{8\text{k}}{17\text{k}+8\text{k}} \times 2.5 = \mathbf{800 \text{ mV}}$
2. Collector current: $I_C = 10^{-17} \exp(800/26) = \mathbf{231 \text{ }\mu\text{A}}$
3. Base current: $I_B = 231 \text{ }\mu\text{A}/100 = \mathbf{2.31 \text{ }\mu\text{A}}$
4. Is I_B negligible? Divider current $= V_{CC}/(R_1 + R_2) = 2.5/(25\text{k}) = 100 \text{ }\mu\text{A}$. Since $2.31 \text{ }\mu\text{A} \ll 100 \text{ }\mu\text{A}$, **yes, I_B is negligible.**
5. Collector voltage: $V_{CE} = 2.5 - (231 \text{ }\mu\text{A})(5\text{k}) = 2.5 - 1.155 = \mathbf{1.345 \text{ V}}$
6. Active mode: $V_{CE} = 1.345 \text{ V} > V_{BE} = 0.8 \text{ V}$. **Confirmed.**

EXAM TIP

Check: Always verify the “negligible I_B ” assumption after solving. Compare I_B to the divider current $I_{\text{div}} = V_{CC}/(R_1 + R_2)$. If $I_B < I_{\text{div}}/10$, the assumption is valid.

5.3 Exact Analysis: Thevenin Equivalent

When I_B is *not* negligible, replace the voltage divider with its Thevenin equivalent:

5.2 Operating Point Analysis and Design 183

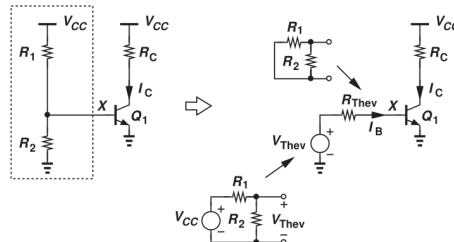


Figure 5.17 Use of Thevenin equivalent to calculate bias.

Moreover, R_{out} is given by the output resistance of the network if V_{CC} is set to zero.

Figure 12: Figure 5.17 from Razavi — Replacing the voltage divider with Thevenin equivalent for exact bias analysis.

DERIVATION: Thevenin Bias Analysis

Thevenin voltage: $V_{\text{Thev}} = \frac{R_2}{R_1 + R_2} V_{CC}$

Thevenin resistance: $R_{\text{Thev}} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$

The base-emitter KVL becomes:

$$V_X = V_{\text{Thev}} - I_B R_{\text{Thev}}$$

$$I_C = I_S \exp\left(\frac{V_{\text{Thev}} - I_B R_{\text{Thev}}}{V_T}\right)$$

Combined with $I_C = \beta I_B$, this system is solved iteratively:

1. Guess $V_{BE} = V_T \ln(I_C/I_S)$ (start with 800 mV)
2. Compute $I_B = (V_{\text{Thev}} - V_{BE})/R_{\text{Thev}}$, then $I_C = \beta I_B$
3. Update $V_{BE} = V_T \ln(I_C/I_S)$
4. Repeat until convergence (usually 2–3 iterations)

5.4 Technique 3: Emitter Degeneration (The Best Approach)

The gold standard for biasing adds a resistor R_E in series with the emitter:

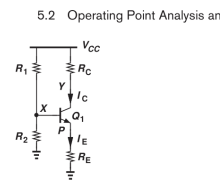


Figure 5.19 Addition of degeneration resistor to stabilize bias point.
vielding

Figure 13: Figure 5.19 from Razavi — Voltage divider bias with emitter degeneration resistor R_E . This is the most robust biasing topology.

DERIVATION: Emitter Degeneration Bias

If I_B is negligible, the voltage at the base is $V_X = \frac{R_2}{R_1 + R_2} V_{CC}$.

The voltage at the emitter (node P) is $V_P = V_X - V_{BE}$.

Since I_E flows through R_E :

$$I_E = \frac{V_P}{R_E} = \frac{V_X - V_{BE}}{R_E}$$

Since $I_E \approx I_C$ (for large β):

$$I_C \approx \frac{1}{R_E} \left(\frac{R_2}{R_1 + R_2} V_{CC} - V_{BE} \right)$$

KEY RESULT:

Why this is superior: I_C depends on R_1 , R_2 , R_E , V_{CC} , and V_{BE} — but **NOT on β** ! Furthermore, if $V_X \gg V_{BE}$, then I_C is also insensitive to V_{BE} variations (temperature).

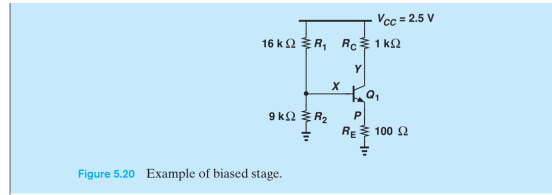
Design rule: Choose V_X to be at least 3–4 times V_{BE} ($\sim 2\text{--}3\text{ V}$) for best stability.

LECTURE INSIGHT (Lecture 20)

Razavi in Lecture 20: “This is the biasing topology you will use for the rest of this course. The emitter resistor provides *negative feedback*: if I_C tries to increase (due to temperature or β change), the voltage drop $I_E R_E$ increases, which *reduces* V_{BE} , which brings I_C back down. The circuit corrects itself. This is beautiful engineering.”

Example 5.10: Biasing with Emitter Degeneration (Razavi)

Problem: Calculate the bias currents in Fig. 5.20 and verify active mode. $\beta = 100$, $I_S = 5 \times 10^{-17}$ A. Determine how much I_C changes if R_2 is 1% higher than nominal.



Solution We neglect the base current and write

$$V_X = V_{CC} \frac{R_2}{R_1 + R_2} = 900 \text{ mV.}$$

Figure 5.20 from Razavi — $R_1 = 16 \text{ k}\Omega$, $R_2 = 9 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, $R_E = 100 \Omega$, $V_{CC} = 2.5 \text{ V}$

Solution:

1. $V_X = \frac{9\text{k}}{16\text{k}+9\text{k}} \times 2.5 = \mathbf{900 \text{ mV}}$
2. Using $V_{BE} = 800 \text{ mV}$ as initial guess: $V_P = 900 - 800 = 100 \text{ mV}$
3. $I_E = 100 \text{ mV} / 100 \Omega = \mathbf{1 \text{ mA}} \approx I_C$
4. Verify: $V_{BE} = V_T \ln(I_C / I_S) = 26\text{m} \times \ln(10^{-3} / 5 \times 10^{-17}) = 26 \times 30.6 = 796 \text{ mV} \approx 800 \text{ mV}$. ✓
5. $V_Y = V_{CC} - I_C R_C = 2.5 - 1\text{m} \times 1\text{k} = \mathbf{1.5 \text{ V}}$
6. Check: $V_{CE} = V_Y - V_P = 1.5 - 0.1 = 1.4 \text{ V} > V_{BE}$. **Active.**
7. **Sensitivity:** If R_2 increases by 1% $\rightarrow R_2 = 9.09 \text{ k}\Omega$
 $V_X = \frac{9.09\text{k}}{25.09\text{k}} \times 2.5 = 905.6 \text{ mV}$ (+5.6 mV change)
 $V_P = 905.6 - 800 = 105.6 \text{ mV}$
 $I_C \approx 105.6\text{mV} / 100 \Omega = 1.056 \text{ mA}$ (+5.6% change)
With degeneration, a 1% resistor error causes only ~5.6% current change.

5.5 Summary of Biasing Topologies

5.6 PNP Biasing

The same biasing principles apply to PNP transistors with reversed polarities. The supply connects to the *emitter* (top), and the collector resistor connects to ground:

EXAM TIP

For PNP biasing, use the same analysis but with V_{EB} replacing V_{BE} :

$$I_B R_B + V_{EB} = V_{CC} \implies I_B = (V_{CC} - V_{EB}) / R_B$$

And verify active mode: $V_C < V_B$ (collector must be *below* base for PNP).

5.2 Operating Point Analysis and Design 191

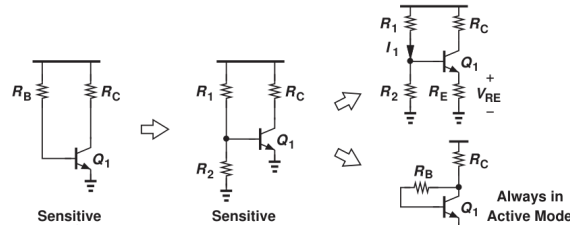


Figure 14: Figure 5.23 from Razavi — Summary: (left) Simple R_B bias — sensitive to β ; (center) Voltage divider — sensitive to resistor errors; (right) With R_E and self-biasing — always in active mode, robust.

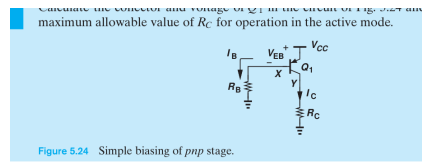


Figure 5.24 Simple biasing of pnp stage.
on The topology is the same as that in Fig. 5.13 and we have $I_B R_B + V_{EB} = V_{CC}$.

Figure 15: Figure 5.24 from Razavi — Simple biasing of a PNP stage. Note: V_{CC} connects to the emitter side, and the collector resistor R_C goes to ground.

6 Practice Problems

Problem 1: Saturation Boundary

An NPN transistor has $I_S = 3 \times 10^{-16}$ A, $\beta = 120$, $V_{CC} = 3$ V, $R_C = 2$ k Ω . It is biased with $V_{BE} = 750$ mV.

- Find I_C .
- Find V_{CE} and verify active mode.
- What is the maximum R_C to avoid saturation (use $V_{BC} = 0$ as the boundary)?

Solution:

- $I_C = 3 \times 10^{-16} \exp(750/26) = 3 \times 10^{-16} \times 3.24 \times 10^{12} = \mathbf{0.972}$ mA
- $V_{CE} = 3 - 0.972\text{m} \times 2\text{k} = 3 - 1.944 = \mathbf{1.056}$ V. Since $1.056 > 0.75$, **active mode**.
- At edge: $V_{CE} = V_{BE} = 0.75$ V.
 $V_{CC} - I_C R_{C,\max} = 0.75 \Rightarrow R_{C,\max} = \frac{3-0.75}{0.972\text{m}} = \mathbf{2.315}$ k Ω

Problem 2: PNP Analysis

A PNP transistor has $I_S = 10^{-16}$ A, $\beta = 80$, $V_A = \infty$. It is connected with $V_{CC} = 2.5$ V at the emitter, base at 1.7 V, $R_C = 500$ Ω to ground.

- Find all terminal currents.
- Find the collector voltage and verify active mode.
- Compute g_m and r_π .

Solution:

- $V_{EB} = 2.5 - 1.7 = 0.8$ V
 $I_C = 10^{-16} \exp(800/26) = \mathbf{2.31}$ mA

$$I_B = 2.31/80 = \mathbf{28.8 \mu A}, \quad I_E = 2.31 + 0.0288 = \mathbf{2.34 \text{ mA}}$$

$$(b) V_C = R_C I_C = 500 \times 2.31\text{m} = 1.155 \text{ V. Since } V_C = 1.155 < V_B = 1.7, \text{ active mode.}$$

$$(c) g_m = 2.31\text{m}/26\text{m} = \mathbf{88.8 \text{ mA/V}}, \quad r_\pi = 80/88.8\text{mS} = \mathbf{901 \Omega}$$

Problem 3: Voltage Divider Bias Design

Design a voltage divider bias network (R_1 , R_2 , R_C , R_E) to achieve $I_C = 1 \text{ mA}$ and $V_{CE} = 1.5 \text{ V}$ with $V_{CC} = 3.3 \text{ V}$. Assume $\beta = 100$, $V_{BE} = 0.8 \text{ V}$.

Solution:

Choose R_E : Want $V_E = R_E I_E \approx R_E I_C$. For good stability, set $V_E \approx 0.5 \text{ V} \Rightarrow R_E = 500 \Omega$.

$$V_X = V_{BE} + V_E = 0.8 + 0.5 = 1.3 \text{ V}$$

$$V_Y = V_E + V_{CE} = 0.5 + 1.5 = 2.0 \text{ V}$$

$$R_C = (V_{CC} - V_Y)/I_C = (3.3 - 2.0)/1\text{m} = \mathbf{1.3 \text{ k}\Omega}$$

$$\text{For divider: } V_X = \frac{R_2}{R_1 + R_2} V_{CC} = 1.3 \Rightarrow R_2/(R_1 + R_2) = 1.3/3.3 = 0.394$$

$$\text{Choose divider current } \approx 10I_B = 10 \times 10 \mu\text{A} = 100 \mu\text{A}$$

$$R_1 + R_2 = V_{CC}/I_{\text{div}} = 3.3/100\mu = 33 \text{ k}\Omega$$

$$R_2 = 0.394 \times 33\text{k} = \mathbf{13 \text{ k}\Omega}, \quad R_1 = 33\text{k} - 13\text{k} = \mathbf{20 \text{ k}\Omega}$$

7 Common Mistakes and How to Avoid Them

COMMON MISTAKES

- Mistake:** Using V_{BE} for a PNP transistor.
Fix: PNP uses V_{EB} (emitter-base). $V_{BE} < 0$ for an active PNP, so using V_{BE} in the exponential gives the wrong sign. Always write $I_C = I_S \exp(V_{EB}/V_T)$.
- Mistake:** Checking active mode with “ $V_C > V_B$ ” for PNP.
Fix: For PNP, the active mode condition is $V_C < V_B$ (collector must be *below* the base voltage). The rule “collector must be more positive than base” is NPN-only.
- Mistake:** Assuming the PNP small-signal model is different from NPN.
Fix: The small-signal models are **identical**. Same r_π , same $g_m v_\pi$, same r_O . All the PNP complexity is in the DC analysis.
- Mistake:** Using $I_C = I_S e^{V_{BE}/V_T}$ when the transistor is in deep saturation.
Fix: In deep saturation, the transistor is an ON switch with $V_{CE,sat} \approx 0.2$ V. The collector current is determined by the *external circuit*: $I_C = (V_{CC} - V_{CE,sat})/R_C$, *not* by the exponential equation.
- Mistake:** Designing a bias circuit that depends on β .
Fix: β varies from device to device (50–200+). Any circuit where $I_C \propto \beta$ will have unpredictable performance. Use voltage divider + emitter degeneration to make I_C depend on resistor ratios instead.
- Mistake:** Forgetting to check the “negligible I_B ” assumption in divider bias.
Fix: After computing I_B , compare it to $I_{div} = V_{CC}/(R_1 + R_2)$. If $I_B > I_{div}/10$, use the Thevenin equivalent for exact analysis.
- Mistake:** Mixing DC and small-signal quantities.
Fix: **Step 1:** Solve DC completely (find I_C). **Step 2:** Compute g_m , r_π , r_O . **Step 3:** Replace transistor with small-signal model, short V_{CC} to ground, and analyze the signal path. Never add I_C and $g_m v_\pi$ in the same equation.

Chapter 2: End-of-Chapter Summary

Saturation Mode

- **Condition:** Both BE and BC junctions forward-biased ($V_{BC} > 0$, i.e., $V_{CE} < V_{BE}$).
- **Soft saturation:** $V_{BC} < 400$ mV. I_C still approximately $I_S e^{V_{BE}/V_T}$.
- **Deep saturation:** $V_{CE,sat} \approx 200$ mV. Transistor is an ON switch. $\beta_{\text{forced}} \ll \beta$.
- **Design rule:** Ensure $V_{CE} > V_{BE} - 400$ mV to stay out of deep saturation.

PNP Transistor

- **Structure:** p -type emitter and collector, n -type base. Holes are majority carriers.
- **Active mode:** $V_{EB} > 0$ and $V_{CB} > 0$ (collector lower than base).
- **Currents:** $I_C = I_S e^{V_{EB}/V_T}$, $I_B = I_C/\beta$, $I_E = I_C + I_B$.
- **Small-signal model:** Identical to NPN. $g_m = I_C/V_T$, $r_\pi = \beta/g_m$, $r_O = V_A/I_C$.

DC Operating Point Design

- **Goal:** Set a stable I_C that is insensitive to β and temperature.
- **Simple R_B :** $I_C = \beta(V_{CC} - V_{BE})/R_B$. Heavily β -dependent. **Never use for amplifiers.**
- **Voltage divider:** $V_X = R_2 V_{CC}/(R_1 + R_2)$ sets V_{BE} . β -independent if $I_B \ll I_{\text{div}}$.
- **With R_E :** $I_C \approx (V_X - V_{BE})/R_E$. Provides negative feedback. **Most robust topology.**
- **Thevenin:** $V_{Th\text{ev}} = R_2 V_{CC}/(R_1 + R_2)$, $R_{Th\text{ev}} = R_1 \parallel R_2$. Use when I_B is not negligible.

The Separation Principle

- **Always** solve DC first, then small-signal.
- DC uses uppercase (I_C , V_{BE}) and the exponential model.
- Small-signal uses lowercase (i_c , v_π) and the linear model ($g_m v_\pi$).
- In small-signal analysis, V_{CC} is an AC ground.
- **Never mix the two analyses.**